MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS)
FOURTH SEMESTER EXAMINATION
(2024 ADMISION ONWARDS)
MG4DSCMAT204-Mathematical Techniques in Business Analytics

Duration: 2 hrs Maximum Marks: 70

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

Part A
Answer any five questions. Each question carries 2 marks

1. State Lagrange's Interpolation Formula. [K][2]

2. Construct a forward difference table from the following data:

x	2	3	4	5	6
$f(x)$	0.3010	0.4771	0.6021	0.6990	0.7782

[U][1]

3. Introduce finite differences with operators [U][1]

4. What is meant by interpolation? How does it differ from Extrapolation?. [U][1]

5. Find the slope intercept formula of a straight line passing through $(2, 3)$ and $(-1, -4)$. [A][3]

6. Solve $\frac{8}{x} + \frac{6}{x+5} = \frac{12}{x}$. [E][3]

7. Find the additional total revenue derived from increasing daily sales from 5 to 9 units, given $MR = 270 - 8x$. [A][4]

8. A producer's marginal cost is MC= What is the total cost TC of producing 2 extra units if 6 units are currently being produced?. [A][4]

Part B
Answer any five questions. Each question carries 6 marks

9. The following results are given:

$\sqrt[3]{27} = 3.0000$	$\sqrt[3]{28} = 3.0369$	$\sqrt[3]{29} = 3.0727$	$\sqrt[3]{30} = 3.1074$

Using them, find $\sqrt[3]{26}$. [A][1]

10. Find the missing term in the following table:

x	0	1	2	3	4
y	1	3	9	...	81

[An][1]

11. What is meant by 'Inverse Interpolation' ? Write down Lagrange's Inverse Interpolation Formula. [U][2]

12. A firm has a fixed cost of 7000 for plant and equipment and a variable cost of 600 for each unit produced. What is the total cost C of producing (a) 15 and (b) 30 units of output?. [An][3]

13. Graph the following quadratic function $f(x) = x^2 - 2$ and identify the vertex and axis. [A][3]

14. Use the point-slope formula to derive the equation for each of the following straight lines and draw the graphs:

    (a) Passing through $(2, 6)$, slope $= 7$

    (b) Passing through $(8, -5)$, slope $= 3$

[A][3]

15. Find the producers' surplus PS for the supply curve $P = Q^2 + 4Q + 60$ at $Q_0 = 5, P_0 = 85$. [E][4]

16. Find the consumers' surplus CS for the demand curve $P = 315 - 3Q^2$ at $Q_0 = 10, P_0 = 75$. [E][4]

Part C
Answer any three questions. Each question carries 10 marks

17. (a) Explain Newton's Forward Interpolation [U][1]

    (b) Using Newton's Forward Interpolation Formula, find the value of $f(x)$ for $x = 3.5$ from the following data:

x	3	4	5	6
$f(x)$	4	13	26	43

[A][1]

18. (a) Show that the third differences of a cubic function are constants. [An][1]

    (b) Using Newton's backward interpolation formula, find the value of the function $f(x)$ when $x = 7$ from the following data:

x	2	4	6	8
$f(x)$	5	18	36	59

[An][1]

19. (a) What is meant by 'Inverse Interpolation' ? Write down Lagrange's Inverse Interpolation Formula. [U][2]

    (b) Given the table of values:

x	35.0	35.5	39.5	40.5
$f(x)$	1175	1280	2180	2420

    Obtain a value of $f(40)$.. [S][2]

20. Using polynomial method find the missing value of the following table,

x	10	20	30	40	50
$f(x)$	15	28	39	...	57

[A][2]

21. (a) A plumber charges a fixed fee of \$50 for a house visit plus an hourly rate of \$35 for every hour of work. Let x be the number of hours and C be the total cost.

    i. Write the linear function $C(x)$ that represents the plumber's total charge. [A][3]

    ii. If the plumber works for 4.5 hours, what is the total cost of the repair? [A][3]

    iii. Suppose the plumber increases the hourly rate to \$45 but keeps the visit fee the same. How does the slope and the equation of the line change? [An][3]

    (b) Explain the nature of the graph of the quadratic function $y = x^2 + 6x + 7$ [U][3]

22. Use the Lagrange multiplier method to optimize the function $z = 6x^2 + 5xy + 2y$ subject to the constraint $2x + y = 96$. [E][4]